3.5

ADULTS

Ear, Nose, and Throat Infections

CHAPTER 3.5	EAR, NOSE, AND THROAT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Otitis Externa			
Acute diffuse (swimmer's ear) *Staphylococcus aureus* *Pseudomonas aeruginosa* Other gram-negatives Polymicrobial Rarely: *Aspergillus spp* *Candida spp* (due to prolonged antibiotic use)	Ofloxacin 0.3% ear drop 10 drops q24H for 7 days ± Steroid ear drops*		• In diffuse otitis externa, systemic antibiotics are not indicated in the absence of fever, spread of infection to pinna or folliculitis. • Aural toilet and keep external ear canal dry. Systemic antibiotics may be needed in severely immunocompromised patients (refer to acute localized otitis external). • If fungal infection is suspected, use ear drops that contain antifungal. • *Not available in UMMC
Acute localised	Ofloxacin 0.3% ear drop 10 drops q24H for 7	**Penicillin Allergy:** Clindamycin 300 mg PO	• Surgical drainage or debridement is usually required.

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<u>Tympanic membrane intact</u> **Mild** Minor discomfort Pruritis Minimal canal oedema **Moderate** Intermediate pain Pruritis Canal partially occluded	days ± Steroid ear drops*	q6-8H for 5 days	• Culture from the ear drainage must be performed ideally before antimicrobial therapy is initiated. • Antibiotics are only indicated in acute localised otitis externa and malignant / necrotising otitis externa. • If systemic symptoms present, may consider Co-amoxiclav 1.2 g IV q8H. • *Not available in UMMC.
<u>Tympanic membrane not intact</u> *Staphylococcus aureus* *Streptococcus pyogenes* *Pseudomonas aeruginosa*	Ofloxacin 0.3% ear drop 10 drops q24H for 7 days ± Steroid ear drops* + Ciprofloxacin 500 mg PO q12H **OR** Co-amoxiclav 625 mg	**Penicillin Allergy:** Clindamycin 300 mg PO q6-8H for 5 days	

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	PO q8H		
Malignant *Pseudomonas aeruginosa*	Ceftazidime 2 g IV q8H	Piperacillin/Tazobactam 4.5 g IV q6H **OR** Cefepime 2 g IV q8H	• Consider switching to Ciprofloxacin PO 750 mg q12H once the patient shows clinical response. • If patient is allergic to Ceftazidime, to refer ID team • Total duration: minimum 4-6 weeks. • Duration is suggested and can be changed depending on clinical response.
Otitis Media			
Viral	**No antimicrobial is required.**		
Bacterial *H influenzae* *Moraxella catarrhalis* *Streptococcus pneumoniae*	Amoxicillin 1 g PO q8H for 5-7 days **If received amoxicillin during previous**	**Penicillin Allergy:** Azithromycin 500 mg PO q24H for 5 days	• Total duration: 5-7 days. • Duration is suggested and can be changed depending on clinical response.

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	consultation, consider treatment with: Co-amoxiclav 625 mg PO q8H for 5-7 days		
Otitis media with suspected intracranial extension	Ceftriaxone 2 g IV q12H + Metronidazole 500 mg IV q8H		• Cases should be discussed between ENT and ID teams.
Chronic suppurative otitis media (CSOM) *P aeruginosa* *S aureus*	Ofloxacin 0.3% ear drop 10 drops q24H for 10-14 days ± Steroid ear drops*		• Aural toilet for discharging ears and keep the external ear canal dry.
Mastoiditis			
Acute **First episode of acute mastoiditis**	Co-amoxiclav 1.2 g IV q8H		• Obtain specimens (deep pus in the ear canal) for C&S. Deescalate/streamline antibiotics once C&S result available • Surgical debridement/drainage if mastoid abscess.

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Streptococcus pneumoniae *Streptococcus pyogenes* *Staphylococcus aureus* *Haemophilus influenzae* **Second episode of acute mastoiditis** *Pseudomonas aeruginosa* Acute: generally, too ill for outpatient therapy	Cefepime 2 g IV q8H	Piperacillin/Tazobactam 4.5 g IV q6H **Penicillin Allergy:** Refer ID team.	• Neurosurgical referral if suspicion of intracranial complications. • Total duration: 7-14 days. Duration is suggested and can be changed depending on clinical response. • If Mastoiditis secondary to chronic otitis media, surgical debridement of auditory canal, then: • Vancomycin 15 mg/kg (max 1 g) IV q8-12H (High risk of MRSA) + Piperacillin/Tazobactam 4.5 g IV q6H
Chronic (associated CSOM) <u>Often polymicrobial:</u> Anaerobes *Staphylococcus aureus* Enterobacteriaceae *Pseudomonas aeruginosa*	Good care of external auditory canal, ear drainage and culture and topical Ofloxacin 10 drops q12H for 2 weeks, is first line. If elective mastoidectomy is indicated, start a brief course of antibiotics based on culture results. If cultures are negative, start empirical antibiotics with:		• Surgical evaluation is indicated for cholesteatoma or consideration for mastoidectomy. • Indications for mastoidectomy: • Chronic drainage and evidence of osteomyelitis by MRI/CT • Evidence of spread to CNS (Epidural abscess, suppurative phlebitis, brain abscess) • Duration: Change to oral once clinically improved. Total
	Piperacillin/Tazobactam 4.5 g IV q6H	Cefepime 2 g IV q8H + Metronidazole 500 mg	

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		IV q8H	duration is 4-6 weeks (usual treatment for osteomyelitis).
Acute Pharyngitis/Tonsillitis			
Viral *Epstein Barr Virus*	No antimicrobial indicated		
Bacterial (less common) *S pyogenes*	Phenoxymethylpenicillin 500 mg PO q8-12H for 7-10 days **OR** Amoxicillin 500 mg PO q8H for 7-10 days **OR** Cefalexin 500 mg PO q8H for 10 days **OR** Benzylpenicillin 2MU IV q6H	**Penicillin Allergy:** Azithromycin 500 mg PO q24H for 3-5 days	• Antibiotics are only required if modified Centor score ≥ 3 or proven bacterial infections: www.mdcalc.com/calc/104/centor-score-modified-mcisaac-strep-pharyngitis. • Total duration: 5 days. • Duration is suggested and can be changed depending on clinical response • If *Streptococcus pyogenes* is suspected, patient maybe has high risk* of rheumatic fever. Refer to Chapter 3.2: Cardiovascular System Infections for management of acute rheumatic fever. • *High risks e.g., existing rheumatic heart disease, scarlet fever.
Diphtheria **Refer to Guideline on Management of Patients**	Diphtheria Antitoxin + Benzylpenicillin 50,000	**If patient can take orally:** Diphtheria Antitoxin	• Anti-toxin is the primary treatment. • Antibiotics are not a substitute for anti-toxin.

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with Suspected/Confirmed Diphtheria Case In UMMC (Available on MyUMMC Staff Portal)	units/kg (max 1.2 MU) IV q12H **Followed by** Phenoxymethylpenicillin 250 mg PO q6H	+ Erythromycin Ethyl Succinate 400 mg PO q6H **OR** 800 mg PO q12H **OR** Phenoxymethylpenicillin 250 mg PO q6H	• IV antibiotics should be switched to oral once clinical improvement occurs. • Total duration: 14 days
Peritonsillar Abscess (Quinsy) *Streptococcus pyogenes* *Streptococcus spp* Other gram-positive aerobes Anaerobes	Co-amoxiclav 1.2g IV q8H	Benzylpenicillin 2MU IV q6H **OR** Ceftriaxone 2 g IV q24H + Metronidazole 500 mg IV q6H **Able to tolerate orally:** Phenoxymethylpenicillin 500 mg PO q6H +	• Surgical drainage is mandatory for abscess. • Obtain specimens for C&S. De-escalate/streamline antibiotics once C&S result available. • IV antibiotics should be switched to oral once clinical improvement occurs. • Total duration 10-14 days. Depending on clinical improvement.

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		Metronidazole 500 mg PO q6H **Penicillin Allergy:** Clindamycin 600 mg IV q8H	
Acute Epiglottitis (Supraglottitis)			
Streptococcus pyogenes *Streptococcus pneumoniae* *Staphylococcus aureus* *Haemophilus influenzae* *Moraxella catarrhalis* Viruses e.g.: *Epstein barr, herpes simplex, influenza* *Parainfluenza*	Co-amoxiclav 1.2 g IV q8H	**Penicillin Allergy:** Ciprofloxacin 400 mg IV q12H + Metronidazole 500 mg IV q8H ± Vancomycin* 15 mg/kg (max 1 g) IV q8-12H	• IV antibiotics should be switched to oral once clinical improvement occurs and patient able to tolerate orally • Total duration: 5 days. Duration is suggested and can be changed depending on clinical response. • If suspected concomitant meningitis, consider Ceftriaxone 2 g q12H. • *To consider Vancomycin if MRSA is suspected: • moderate to severe sepsis • concomitant meningitis • from area with high prevalence of MRSA
Parapharyngeal Space Infection			

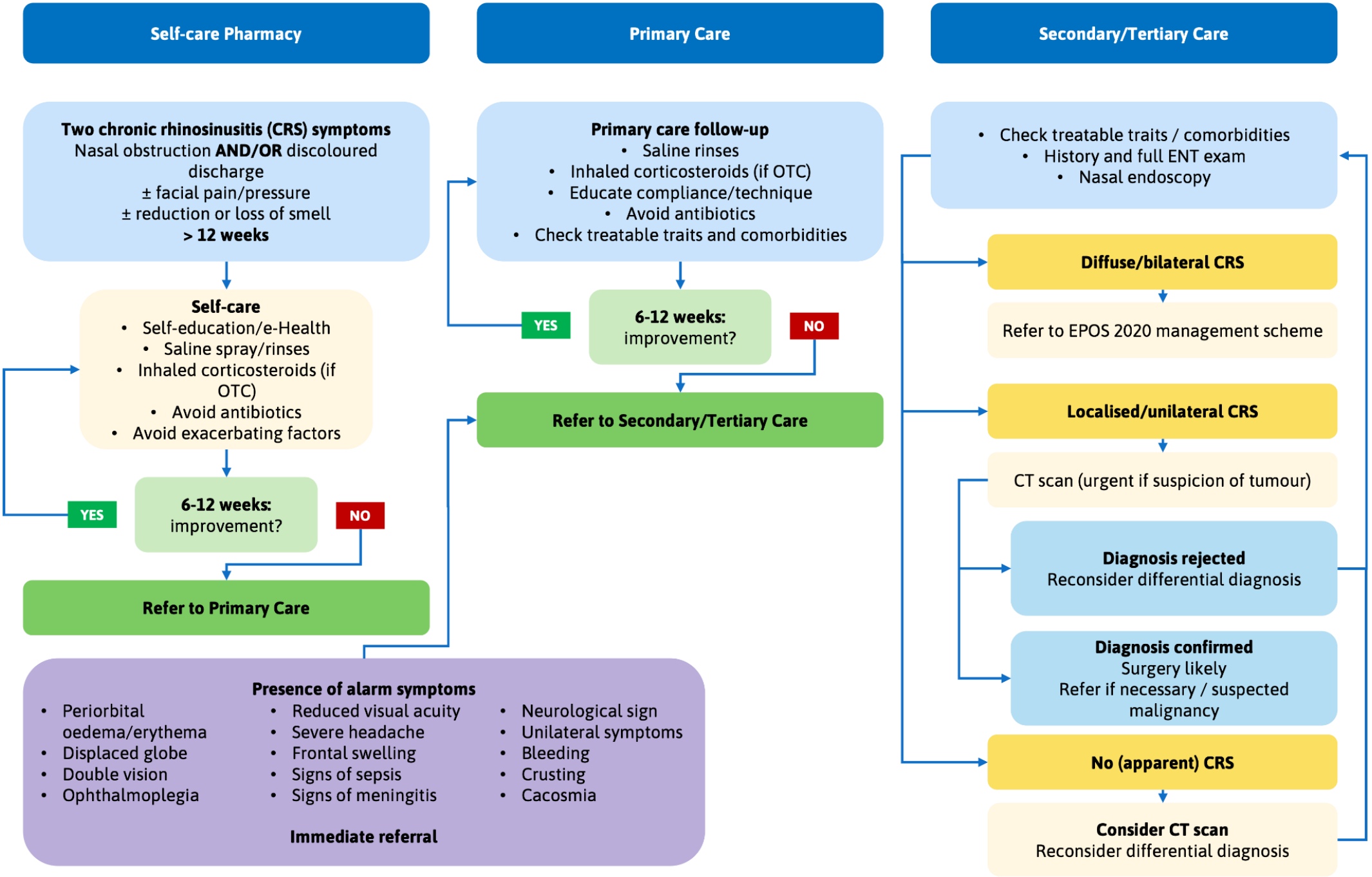
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Parapharyngeal Abscess (Deep neck abscess) Empirical Polymicrobial: Mostly Anaerobes *Streptococcus spp* *Staphylococcus aureus* **Lemierre's syndrome** *Fusobacterium necrophorum*	Co-amoxiclav 1.2 g IV q8H	Cefazolin 2 g IV q8H + Metronidazole 500 mg IV q8H **Penicillin Allergy:** Clindamycin 600-900 mg IV q8H	• Surgical drainage is mandatory for abscess. • Obtain specimens for C&S. De-escalate/streamline antibiotics once C&S result available. • IV antibiotics should be switched to oral once clinical improvement occurs • Total duration: Minimum 2 weeks and until resolution of abscess/ clinical improvement.
Laryngitis			
90% viral	Antimicrobial not indicated		
Acute Rhinosinusitis			
Refer to flowchart from European Position Paper on Rhinosinusitis and Nasal Polyps 2020 (doi: 10.4193/Rhin20.600.).			
Acute Mostly viral			• Avoid routine use of antibiotics for ARS.

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Streptococcus pneumoniae *Haemophilus influenzae* *Moraxella catarrhalis* *Staphylococcus aureus*			• Uncomplicated ARS resolves within 7-10 days without antibiotic treatment.
Mild	Amoxicillin 500 mg PO q8H **OR** Cefuroxime 500 mg PO q12H **If recent antibiotic use:** Co-amoxiclav 625 mg PO q12H	**Penicillin Allergy:** Doxycycline 100 mg PO q12H (avoid in pregnancy) **OR** Azithromycin 500 mg PO q24H (preferred in pregnancy)	• Consider antibiotics if present at least 3 of below: • Purulent/ greenish nasal discharge • Severe local pain (VAS 8-10) • Fever • Elevated ESR/CRP • Double sickening (becoming worse after initial recovery) • VAS: Visual Analogue Score. • The patient is asked: “How troublesome are your symptoms?” • Not troublesome (0) to the worst thinkable troublesome (10). • Suggested duration of therapy is 5 days. However, longer duration may be required depending on clinical response. • Consider diagnostic tap/aspiration if treated with antibiotics.
Moderate/Severe	Co-amoxiclav 1.2 g IV q8H **OR** Cefuroxime 1.5 g IV q8H	**Penicillin Allergy:** Azithromycin 500 mg IV q24H	

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			Send for culture.

Care Pathways for Chronic Rhinosinusitis

Adapted from Fokkens, W. J., et al. **"Executive summary of EPOS 2020 including integrated care pathways." Rhinology (2020).**


Self-care Pharmacy
Primary Care
Secondary/Tertiary Care
Two chronic rhinosinusitis (CRS) symptoms
Nasal obstruction AND/OR discoloured discharge
± facial pain/pressure
± reduction or loss of smell
> 12 weeks
Self-care
• Self-education/e-Health
• Saline spray/rinses
• Inhaled corticosteroids (if OTC)
• Avoid antibiotics
• Avoid exacerbating factors
6-12 weeks: improvement?
YES
NO
Refer to Primary Care
Primary care follow-up
• Saline rinses
• Inhaled corticosteroids (if OTC)
• Educate compliance/technique
• Avoid antibiotics
• Check treatable traits and comorbidities
6-12 weeks: improvement?
YES
NO
Refer to Secondary/Tertiary Care
Presence of alarm symptoms
• Periorbital oedema/erythema
• Displaced globe
• Double vision
• Ophthalmoplegia
• Reduced visual acuity
• Severe headache
• Frontal swelling
• Signs of sepsis
• Signs of meningitis
• Neurological sign
• Unilateral symptoms
• Bleeding
• Crusting
• Cacosmia
Immediate referral
• Check treatable traits / comorbidities
• History and full ENT exam
• Nasal endoscopy
Diffuse/bilateral CRS
Refer to EPOS 2020 management scheme
Localised/unilateral CRS
CT scan (urgent if suspicion of tumour)
Diagnosis rejected
Reconsider differential diagnosis
Diagnosis confirmed
Surgery likely
Refer if necessary / suspected malignancy
No (apparent) CRS
Consider CT scan
Reconsider differential diagnosis

References:

1. Goguen LA et. al. External otitis in adults: Treatment. In: UpToDate, Connor RF (Ed), Wolters Kluwer. (Accessed on May 18, 2024)
2. Limb CJ et. al. Acute otitis media in adults. In: UpToDate, Connor RF (Ed), Wolters Kluwer. (Accessed on May 18, 2024)
3. Guidelines for the Treatment of Ear, Nose and Throat Infections in Adults. University Hospitals of Leicester (NHS), 24th April 2020.
4. The Sanford Guide to Antimicrobial Therapy 2024. (Accessed on May 18, 2024)